



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Midterm Exam, Trimester: Spring 2024

Course Code: CSE-1115, Course Title: Object Oriented Programming

Total Marks: 30, Duration: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

QUESTION 1

[3+2+2.5 MARKS]

Consider the following codes:

```
class Point2D
{
    int x, y;
    public Point2D(int x, int y){
        this.x = x;
        this.y = y;
        System.out.println("Point2D constructor");
    }
    public String Display(){
        //write codes here
    }
}
class Point3D extends Point2D{
    int z;
    //write codes here
}
```

```
public class Test{
    public static void main(String args[]){
        Point2D p2D = new Point2D(1, 2);
        System.out.println(p2D.Display());
        Point3D p3D = new Point3D(5, 4, 3);
        System.out.println(p3D.Display());
    }
}
```

Now:

- I. Complete the "Display" method of the Point2D class that prints all the instance variables,
- II. Add a constructor in the Point3D class that uses the base class constructor,
- III. Add another method "Display" in the Point3D class. You have to use the parent's "Display" method here, so that the **output** looks like this:

```
Point2D constructor ✓
x= 1, y= 2 ✓
Point2D constructor ✓
Point3D constructor ✓
x= 5, y= 4, z= 3 ✓
```


QUESTION 2

Modify the following program by including/excluding the some codes without changing the highlighted parts. [3+2+1.5+1 MARKS]

```
public class Myparent {
    private int p;
    public final int myfunction(){
        return p*p;
    }
    public void set_p(int Q){p = Q;}
    // Write your code here
}

public class Mychild extends Myparent{
    public Mychild(int K){ super(K); }

    public final int myfunction(){
        return p*p+1;
    }
    // write your code of myroot() that finds the square
    // root of p in class Myparent
    // write other necessary codes here
}
```

```
public class Mytest {
    public static void main(String[] args) {
        Myparent c1, c2;
        c1 = new Mychild(2);
        c2 = new Mychild();
        c2.set_p(2);
        int x = c2.myfunction();
        double y = ((Mychild) c1).myroot(); // find square
        // root of p in class Myparent
        System.out.println("x = " + x + ", y = " + y);
    }
}
```

QUESTION 3

Write the output of the following program:

[7.5 MARKS]

```
class Person{
    int id;
    String name;
    static int s = 10;
    {
        System.out.println("3");
    }
    public Person(){
        this.id = 1;
        this.name = "M";
        System.out.println("1");
        s++;
    }
}
```

```
public Person(int id, String name){
    this();
    this.id = id;
    this.name = name;
    System.out.println("2");
    s++;
}

public static void main(String args[]){
    Person p = new Person(1, "N");
    Person p1 = new Person();
    System.out.println(p1.s);
    p.s = 11;
    System.out.println(Person.s);
}
```


QUESTION 4

[7.5 MARKS]

Suppose that you visit a village market where fresh vegetables and fishes are sold. The sellers sell their items with a profit of $z\%$ of their production cost c . Typical items are given by the following Table:

Food Items	Type t	production cost c per Kg	profit of $z\%$
vegetable	Spinach	20	15
vegetable	Cauliflower	25	18
fish	Carp	300	15
fish	medium	250	20
fish	small	200	25

The class FoodItem that includes type t , production cost c and profit z as public variables and a method findprice() is given as follows:

```
public class FoodItem {
    public double c, z;
    public String t;
    public double getprice(double amount){
        return c*amount*(1+z/100);
    }
}
```

Next, two derived classes Vegetable and Fish are given as follows.

```
public class Vegetable extends FoodItem{
    public void setparameter(){
        if(t == "Spinach"){ c = 20; z = 15; }
        else if(t == "Cauliflower"){c = 25; z = 18;}
    }
    public Vegetable(String t){
        this.t = t;
    }
}
```

```
public class Fish extends FoodItem{
    public void setparameter(){
        if(t == "Carp"){ c = 20; z = 15; }
        else if(t == "medium"){c = 25; z = 20;}
        else if(t == "small"){c = 200; z = 25;}
    }
    public Fish(String t){
        this.t = t;
    }
}
```

Now as a programmer, test the above classes in the main method in a new class MyTest by finding your total purchase price if you buy 3Kg fish of type small and 2 Kg vegetable of type Cauliflower. [Make 2 objects of FoodItems and use the child class references to a FoodItems class object. Then call appropriate methods e.g., setparameter, getprice.]